SAP Business Network For Logistics - Road EDI Integration Tips and Tricks



Jeff-Gebo
Product and Topic Expert

2023 Jul 26 10:45 PM

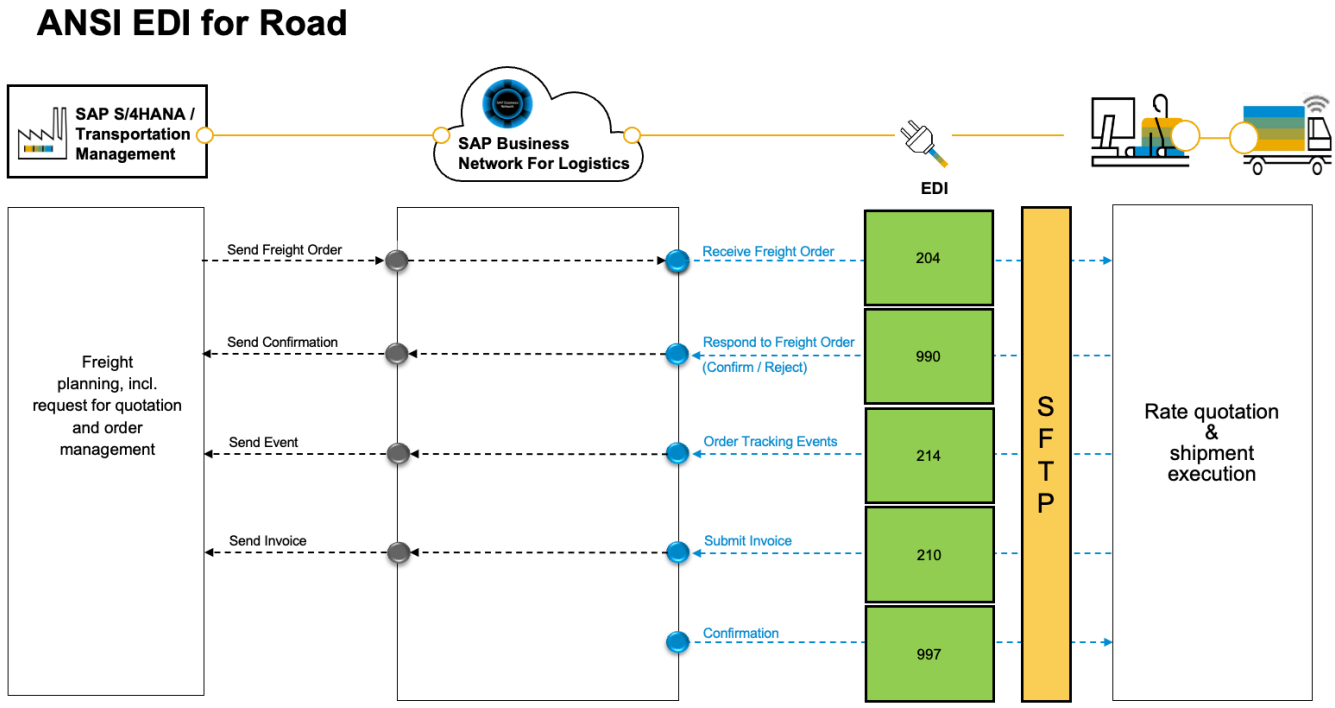
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Introduction

SAP Business Network for Logistics (BN4L) provides collaboration capabilities for shippers and carriers - freight tendering, subcontracting, settlement, tracking and dock appointment scheduling can all be collaborated on. A carrier on our network can connect and transact with any shipper on the network. BN4L comes by default with a web portal that our carriers can use to collaborate with shippers they are connected to. Optionally, carriers can build an integration from BN4L to their own systems using EDI or API. If you haven't yet read my blog on EDI vs API integration I highly recommend doing so.

Over the past year and a half, I have worked with hundreds of carriers who are building these integrations, both EDI and API. Over the course of this work with carriers going the EDI integration route I have identified common issues and questions that they have. My hope is that the below list will assist in speeding up EDI integrations for new carrier integrations. **Note**: *I work with North American carriers so I am focused on ANSI EDI, that said we also support EDIFACT for the rest of the world.*



EDI ANSI Common Issues & Questions

204

The first question I get from a carrier is always this: What are your ISA/GS IDs (AKA sender / receiver IDs)?

In BN4L we use the LBN ID of the shipper and carrier. The LBN ID is a unique ID (10 digit numeric value) of the shipper and carrier the uniquely identifies them on the network. Each will typically have two of these. One for the test landscape and another for the production landscape. Like any good integrator you will first build and test your integration on the test landscape then move to production once everything has been successfully tested. Ok, then there is the second related question – what is the qualifier? Ours is “ZZ”. You can find all of this documented in our 204 Spec Guide. **PRO TIP:** *Never download our guides, just bookmark the link – we are making improvements all the time and upgrade our guides regularly.*

990

The easiest of all the EDI inbound messages but almost everyone misses this one segment, you need to send the Sender System ID! Why? Because a shipper can possibly have multiple systems connected to their BN4L tenant and due to that we need the carrier to send back this segment so we can route the message to the proper system. If only I had a dollar for every time I emailed the below snippet, I could retire. LOL! Learn more in the 990 spec guide.

You are missing the following segment:

N9*06*"Sender System ID"~

You can get the sender system ID value from the 204 segment:

L11*SenderSystemId*06~

210

This one is more complex – the most difficult of all the inbound messages. First you need to send back three N9 segments – the Sender System ID, the Bill From Party and the Purchasing Party. The Bill From Party and Purchase Party are the business partner IDs as maintained in the shippers system. If you don't get these correct the invoice won't process successfully. Sometimes carriers forget to send the C3 segment with the currency code. Most difficult about the 210 is how we require the L1 segments to be formatted. I'm not going to go into detail on that in this blog, I will create a separate one just for L1 segment. Suffice to say, take the advice I mention below about the L1 segment – again, I email this snippet out all the time to carriers trying to integrate with the 210:

You are missing the following segments:

N9*06*Sender System ID~

N9*12*BillFromParty~

N9*CN*PurchasingParty~

You can get these from the 204 in the following segments:

L11*SenderSystemId*06~

L11*PurchasingParty*CN~

L11*BillFromParty*12~

With the 210 please pay special attention to how we format the L1 – look at the examples on page 30 of the 210 spec guide. **PRO TIP:** *You need to ask the shipper you are integrating with which BN4L Charge Codes they support for your use case with them. They should be able to send you that list.*

214 for Freight Collaboration Tracking (not GTT)

This one isn't too bad, but not used quite as much because a lot of our shippers choose a Visibility Provider (like Project44, FourKites or Shippeo to name a few) to get order tracking data. That said, more and more carriers are setting up the 214 with us...I imagine this will become more popular over time. In addition to the Sender System ID we need the Stop ID or Location ID. Stop ID is just 1, 2, 3 (depends on number of stops), or better yet send the Location ID which can be found in the 204. Below is the email snippet I send out regularly to carriers and as always the 214 Spec Guide which has examples is your friend.

You are missing the following segment:

L11*SenderSystemId*06~

You can get the sender system ID value from the 204 segment:

L11*SenderSystemId*06~

You also need to send a Stop ID or Location ID.

The location ID can be found in the 204 N1-04 position.

Send as: L11*Location ID*LU~

997

We currently only support the 997 for confirmation of the files you send to BN4L - these are the 990, 214 and 210. It is on our roadmap to support 997 confirmations for a 204 (hopefully will be delivered sometime this year). With our 997 we have one thing that needs to be known by carriers - that is the sender ISA / GS ID is "LBN". Why? Well because we did receive your file, but it may not have been routed all the way to the shipper's TM. This 997 only means that it was received by BN4L. (Why "LBN" and not "BN4L" well, BN4L used to be called Logistics Business Network, LBN for short).

Summary

EDI Integration can be easy if you consider these common issues and answers to common questions listed here. I've seen this integration setup as quickly as a day...I've also seen it take weeks. My hope is with this info we can get more carriers integrated in days.

Feel free to ask any questions below on this in the comments. Happy integrating!

PS - we now support the 300 and 301 for Ocean. Are you a carrier interested in this? Let me know!

For further questions and feedback related to the blog, please check the Q&A area and feel free to post your questions in the link with the tag SAP Business Network for Logistics.

LABELS

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TAGS

API AND EDI INTEGRATION IN SAP BUSINESS NETWORK FOR LOGISTICS | BN4L | EDI | Integration | Lbn | LBN Connectivity

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Comments

C Chida_Chauhan Explorer 2023 Jul 27 2:51 AM

Thanks Jeff for sharing this information. Based on our experience with the carriers and you,

Yes these were the common issues we have noticed when integrating EDI with carrier using LBN

Thx

Chida

K Kati_WN55 Explorer

2023 Jul 27 7:13 AM

Thanks for sharing, Jeff !



Doug_DeLuca

Product and Topic Expert

2023 Jul 27 1:44 PM

The 214! This is so good to have. Thanks for calling that out.

And thanks for mentioning Ocean at the end. A bit of something different there. May be good to post more about that too.

May also be worth noting that once a carrier has an account on SAP Business Network, they only need to join once, and can integrate as described via EDI or API, or transact via portal, with many shippers, all from that one carrier account. Some shipper *requirements* may differ from shipper to shipper, but the carrier account can stay the same. *"Connect once, transact with many"*

Thank you for this.

S Saurabh Participant

2023 Jul 30 5:09 AM

Hi Jeff

thanks for sharing...

a quick question- as per your flow- event will be reported from BN4L but in real life specially in shipper scenario where EWM is integrated... event are reported from EWM (like arrival at door , departure etc.) and it changes TM execution status but that FO status in TM and FO status on LBN shipper app is not in sync

i mean to ask- how can we keep them in sync



RavishShettyR

2023 Jul 31 6:19 AM

Product and Topic Expert

HI Saurabh,

Ideally events also flow back to TM system. So customers can view the events seen in LBN in TM too. However, the FO status is a derived status and it can be custom defined. on LBN only execution status is tracked.

BR, Ravish.

D

Dave_Nava Discoverer

2023 Oct 30 8:32 PM

Hello Jeff,

Actually we are working on a project that we will had 4 Ocean Carriers (this 4 are on top ten over ocean), this carrier had already confirm us that they are working with EDIFACT (IFTMBF, IFTMBC, IFTMIN, VERMAS, IFTMCS, IFTFCC) and they don't work with ANSI X12 (300, 301, 304, 310).

Also this carrier tell us that they will only work with their owns APIs and EDIFACT. We are performing all this conversion from SAP API (Rest) to the corresponding EDI using SAP Integration Suite, But this will impact the ocean carrier.

Are you planning to on the Roadmap to develop the EDIFACT that we mention before?

Best Regards

David Nava



Jeff-Gebo

Product and Topic Expert

2023 Nov 06 6:31 PM

We do support EDIFACT for Road, see here:

https://help.sap.com/docs/business-network-freight-collaboration?task=integrate_task

As for the rest I will ask.



Jeff-Gebo

2023 Nov 07 6:05 PM

Product and Topic Expert

Hi David,

For ocean EDIFACT support we recommend you put in an influence request in the influence portal. They do have plans to do this but nothing defined on the roadmap yet.

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